BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Firewalls and Network Security

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How does a brute-force attack work?

2. Which medium resists electromagnetic interference?
   A) Copper Ethernet cable
   B) Wi-Fi radio
   C) Bluetooth radio
   D) Fibre optic cable

3. How does wired Ethernet differ from Wi-Fi?

4. Choose links for a nearby wireless keyboard and a high-capacity link between school buildings. Why does Bluetooth suit the keyboard but usually not the buildings?

Today's objectives

Explain the purpose of firewalls and anti-malware software. Describe user access levels and security updates.

Key vocabulary - Read, cover, recall

Vulnerability
Weakness that could be exploited to compromise a system.

Firewall
Monitors/filters network traffic using security rules.

Anti-malware
Software detecting and removing malicious programs.

Security update
Change fixing bugs or security weaknesses.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Vulnerabilities
A vulnerability is a weakness that an attacker might exploit. Using multiple security measures reduces risk.

Firewalls
A firewall monitors/filters network traffic using rules, blocking or allowing connections according to security policy.

Anti-malware and updates
Anti-malware software detects and may remove malicious software. Software updates can fix known security weaknesses.

User access levels
Access rights restrict what users can view or change, reducing the damage possible if a lower-privilege account is compromised.

I Do - Worked model

Explain how a firewall and user access levels help protect a school file server. [4 marks]

- The file server faces network traffic and has confidential resources.
- A firewall filters incoming/outgoing network traffic [1], blocking connections that violate security rules [1].
- User access levels limit permissions [1], preventing ordinary users from editing or viewing protected files [1].
- Two distinct defences with their actions and security benefits.

We Do - Guided practice

1. A classroom PC may be infected by downloaded malware. Explain how anti-malware software and updates reduce risk. [4 marks]

2. A school wants to limit access to staff records. Explain how access levels can help. [4 marks]

3. A school firewall blocks an unfamiliar network connection. Explain what the firewall does and one limitation. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the simulated firewall action.

```
blocked = true
if blocked then
print("Connection blocked")
endif
```

You Do - Independent exam questions

Q2. Define a firewall. [2 marks]

Q3. A public library allows visitors to use shared PCs while storing private customer booking details. Explain how anti-malware software and user access levels can protect the library's systems. [4 marks]

Q4. Explain how firewalls, updates and permissions combine to reduce security risks. [6 marks]

Exit Ticket - True or False?

Installing one firewall guarantees that a computer cannot be infected by malware.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It tries many possible passwords or keys until one works.

2. D - Fibre optic cable

3. Ethernet uses cables. Wi-Fi uses radio, allowing movement but risking interference.

4. Bluetooth suits a nearby keyboard's short-range, low-data needs. Links between buildings need more range and capacity, such as fibre.

I Do - Explain the model

The file server faces network traffic and has confidential resources.

A firewall filters incoming/outgoing network traffic [1], blocking connections that violate security rules [1].

User access levels limit permissions [1], preventing ordinary users from editing or viewing protected files [1].

Two distinct defences with their actions and security benefits.

We Do - Model answers

1. Anti-malware scans for malicious code [1] and can quarantine/remove it [1]. Security updates patch known software weaknesses [1], making exploitation of those vulnerabilities less likely [1].

2. Permissions give access only to authorised accounts [1], restricting who can read records [1]. A normal student account has lower privileges [1], reducing the changes it can make to sensitive information [1].

3. Firewall uses rules to check network traffic [1] and blocks disallowed connections [1]. It cannot guarantee stopping every threat [1], especially if a user runs malware already inside the device [1].

Code First - Answer

```
blocked = true
if blocked then
print("Connection blocked")
endif
```

Expected output: Connection blocked

You Do - Self-mark

Q2: Monitors/filters network traffic [1] according to rules to block unauthorised traffic [1].

Q3: Anti-malware software scans for malicious programs [1] and can quarantine/remove them to reduce harm [1]. User access levels restrict permissions on public accounts [1], preventing visitors from accessing or changing sensitive customer-booking records [1].

Q4: Firewall filters traffic [1], blocking disallowed connections [1]. Updates repair software weaknesses [1], reducing opportunities for exploitation [1]. Access levels restrict users' permissions [1], limiting exposure or damage [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.